Karhunen-Loeve Transform

有关 KLT 必须要掌握其运算过程，14-15-B 的 Question2.b（16 marks），c（5 marks）以及 12-13-C 的 Q2.b（17 marks）就是考察了 KLT 的流程以及一些问题的说明。13-14-A 的 Question2.c（10 marks）让写出 KLT 用到的 orthonormal set of basic functions 是如何得到的，以及说明 KLT 的四个缺点，C 卷在 Questionn2 的选择题（2 marks）内考察了 KLT 的优点。

综上，期末考试复习（预习）时间紧张，建议优先掌握 KLT 的整个运算过程以及 KLT 的优缺点，然后再去掌握一些关于 PCA 等的知识，感觉高级变换考试不会放过任何一个细节。

以下是相关内容：

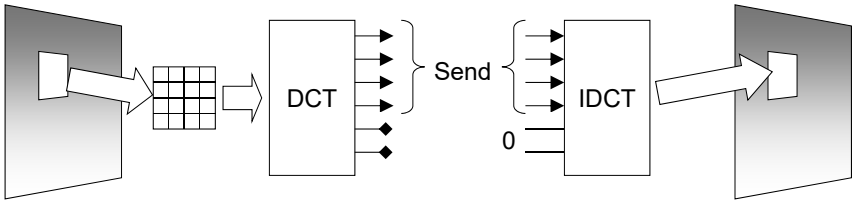
Karhunen Loeve Transform

Met Discrete Cosine Transform (DCT) in a previous lecture.

An advantage of DCT:

- most energy concentrated in a few coefficients, so
- can discard some coeffs, while keep most of signal

E.g. image compression:



Fourier, Wavelets also do this, depending on the signal.

But - what is the “best” transform for this?

Principal Components Analysis (PCA)

- Multivariate procedure
- Main use of PCA is to reduce dimensionality of a data set while retaining as much information as is possible.
- Finds a projection of the observations onto orthogonal axes contained in the space defined by the original variables.
- Correlated variables transformed into uncorrelated variables
  - Ordered by reducing variability.
  - Uncorrelated variables are linear combinations of original variables
- Computes compact, optimal description of data set.
- Rotates data so that maximum variabilities projected onto the axes
- Rotation of existing axes to new positions in the space defined by the original variables.

Covariance Matrix

- Find covariance matrix

$$\mathrm{cov}[\mathbf{X}] = \begin{bmatrix} 7.5649 & 3.8464 \\ 3.8464 & 3.2451 \end{bmatrix}$$

- Find principal components

$$\mathbf{pc} = \begin{bmatrix} 0.8630 & -0.5052 \\ 0.5052 & 0.8630 \end{bmatrix}$$

- Are they orthogonal?

$$\langle \psi_1, \psi_2 \rangle = \langle (0.8630, 0.5052), (-0.5052, 0.8630) \rangle$$

$$= 0.8630 \cdot (-0.5052) + 0.5052 \cdot 0.8630 = 0 \quad \textbf{Yes}$$

- Are they normalised?

$$\langle \psi_1, \psi_1 \rangle = 0.8630 \cdot 0.8630 + 0.5052 \cdot 0.5052 = 1$$

$$\langle \psi_2, \psi_2 \rangle = (-0.5052) \cdot (-0.5052) + 0.8630 \cdot 0.8630 = 1 \quad \textbf{Yes}$$

Now let's give the theory



- The Karhunen Loève Transform is based on PCA

- Also known as the Hotelling Transform

Karhunen Loève Transform (KLT)

- Basis functions are eigenvectors of the covariance matrix R_{XX} of the input signal.
  - This set of basis vectors is *not* fixed
  - Basis vectors depend on the data set
- KLT yields decorrelated transform coefficients (covariance matrix R_{YY} is diagonal).
- Optimal linear transform for keeping the subspace that has largest variance
  - Achieves optimum energy concentration.
- KLT maximizes coding gain

Procedure



1. Find mean vector for input data $\mathbf{X} = [\vec{x}_0, \vec{x}_1, \ldots \vec{x}_{N-1}]$

$$E(\vec{x}) = \frac{1}{N}\sum_{i=0}^{N-1}\vec{x}_i$$

2. Find covariance matrix

$$\mathbf{R}_{\mathbf{XX}} = \frac{1}{N-1}\sum_{i=0}^{N-1}(\vec{x}_i - E(\vec{x}))(\vec{x}_i - E(\vec{x}))^T$$

3. Find eigenvalues of the covariance matrix

$$\left|\mathbf{R}_{\mathbf{XX}} - \lambda\mathbf{I}\right| = 0$$

$|\mathbf{A}|$ means "Determinant of $\mathbf{A}$"

4. Find eigenvectors of the covariance matrix

$$\left[\mathbf{R}_{\mathbf{XX}} - \lambda_i\mathbf{I}\right]\vec{\phi}_i = 0$$

Procedure

5. Normalise the eigenvectors

$$\left\langle \vec{\phi}_i, \vec{\phi}_i \right\rangle = 1$$

6. Transform the input

$$\mathbf{Y} = \phi^T\mathbf{X}$$

7. *To check,* find covariance matrix of $\mathbf{Y}$

$$\mathbf{R}_{\mathbf{YY}} = \frac{1}{N-1}\sum_{i=0}^{N-1}(\vec{y}_i - E[\vec{y}])(\vec{y}_i - E[\vec{y}])^T$$

8. *Optional,*
    1. set last row vector(s) of $\mathbf{Y}$ to 0

$$\mathbf{Y} \rightarrow \mathbf{Y'}$$

    2. Inverse transform this

$$\mathbf{X}' = \phi\mathbf{Y}'$$

Drawbacks of KLT

KLT is theoretically optimal (in the MSE sense). (KLT maximises the coding gain, i.e. maximises the SNR after a given level of compression.)

BUT, it has practical difficulties:

- Estimate of correlation can be unwieldy
- Solution of eigenvector decomposition is computationally intensive (i.e. slow)
- Calculation of forward and inverse transforms is $O(MN)$ for each image block
- Transmission of data-dependent basis $\mathbf{A}$ is required
- The technique is linear, therefore any non-linear correlation between variables will not be captured.

In comparison, turns out that DCT is fixed, and

- a good approximation to KLT for typical images,
- needs no eigenvalue decomposition, and
- transform is $O(N \log N)$.

Wigner-Ville Distribution

WVD 在 14-15 年题目中考察了利用定义公式进行推导证明，如 14-15-B-Question4（25 marks），14-15-C-Question4.b（14 marks）。在 13-14 年题目中多以选择的形式出现，选择主要考查了 WVD 的优缺点与性质，例如 13-14-B-Question2 两道选择题（4 marks），13-14-C-Question3、4 共两道选择（4 marks）。

建议根据个人情况对 WVD 的定义式以及优缺点进行掌握。

以下是相关内容：

Wigner-Ville Distribution

STFT, Wavelet transform etc:

- Compute correlations between signal and basis fns.
- Time-freq resolution determined by basis fns.

Another approach:

- Time-frequency energy density - signal's energy density in both time and freq
- (c.f. power spectrum: energy in freq only)
- e.g. Wigner-Ville distribution

An “instantaneous power spectrum”

Recall the power spectrum :

$$P(\omega) = |S(\omega)|^2 = \int_{-\infty}^{\infty} R(\tau) e^{-j\omega\tau} d\tau$$

where $R(\tau)$ is the autocorrelation function (acf)

$$R(\tau) = \int_{-\infty}^{\infty} s(t) s^*(t-\tau) dt = \int_{-\infty}^{\infty} s(t+\tau/2) s^*(t-\tau/2) dt$$

What happens if we use *instantaneous* autocorrelation :

$$R(t,\tau) = s(t+\tau/2) s^*(t-\tau/2)$$

instead of $R(\tau) = \int_{-\infty}^{\infty} R(t,\tau) dt$? We get :

$$WVD_s(t,\omega) = \int_{-\infty}^{\infty} s(t+\tau/2) s^*(t-\tau/2) e^{-j\omega\tau} d\tau$$

which is the *Wigner - Ville Distribution* (WVD).

Cross-WVD vs auto-WVD

Since we can define a cross - correlation,
we can also define a cross - Wigner - Ville distribution :

$$WVD_{s,g}(t,\omega) = \int_{-\infty}^{\infty} s(t+\tau/2) g^*(t-\tau/2) e^{-j\omega\tau} d\tau$$

Taking complex conjugates we find

$$WVD_{s,g}(t,\omega) = WVD^*_{g,s}(t,\omega)$$

So for the usual WVD ("auto - WVD") we have

$$WVD_s(t,\omega) = WVD_{s,s}(t,\omega) = WVD^*_s(t,\omega)$$

so the auto - WVD is always *real*.

WVD Properties:
Time Marginal Condition

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} WVD_s(t,\omega)d\omega = \int_{-\infty}^{\infty} s\left(t+\frac{\tau}{2}\right)s^*\left(t-\frac{\tau}{2}\right)\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-j\omega\tau}d\omega d\tau$$

$$= \int_{-\infty}^{\infty} s\left(t+\frac{\tau}{2}\right)s^*\left(t-\frac{\tau}{2}\right)\delta(\tau)d\tau$$

$$= |s(t)|^2$$

So intregral over frequency of WVD is the signal power density at time t

Compare similar result for probability densities :

$$p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)dy$$

Frequency Marginal Condition

$$\int_{-\infty}^{\infty} WVD_s(t,\omega)dt = \int_{-\infty}^{\infty} s\left(t+\frac{\tau}{2}\right)s^*\left(t-\frac{\tau}{2}\right)\int_{-\infty}^{\infty} e^{-j\omega\tau}dtd\tau$$

$$= \int_{-\infty}^{\infty} e^{-j\omega\tau}\int_{-\infty}^{\infty} s(t)s^*(t-\tau)dtd\tau$$

$$= \int_{-\infty}^{\infty} e^{-j\omega\tau}R(\tau)d\tau$$

$$= |S(\omega)|^2$$

So intregral over time of WVD is the power spectral density

We also have :

$$\frac{1}{2\pi}\int_{-\infty}^{-\infty}\int_{-\infty}^{\infty} WVD_s(t,\omega)dtd\omega = \frac{1}{2\pi}\int_{-\infty}^{-\infty}|S(\omega)|^2\,d\omega = \int_{-\infty}^{\infty}|s(t)|^2\,dt$$

i.e. the WVD is unitary : the energy in $WVD_s(t,\omega)$ is equal to energy in original signal $s(t)$.

Time-shift & Freq-moduln. invariant

If the WVD of $s(t)$ is $WVD_s(t,\omega)$,
then the WVD of time-shifted signal $s_0(t) = s(t-t_0)$
is a time-shifted WVD : $WVD_{s_0}(t,\omega) = WVD_s(t-t_0,\omega)$

Further, the WVD of frequency-modulated signal
$s_1(t) = s(t)e^{j\omega_1 t}$ is a frequency-shifted WVD :
$WVD_{s_1}(t,\omega) = WVD_s(t,\omega-\omega_1)$

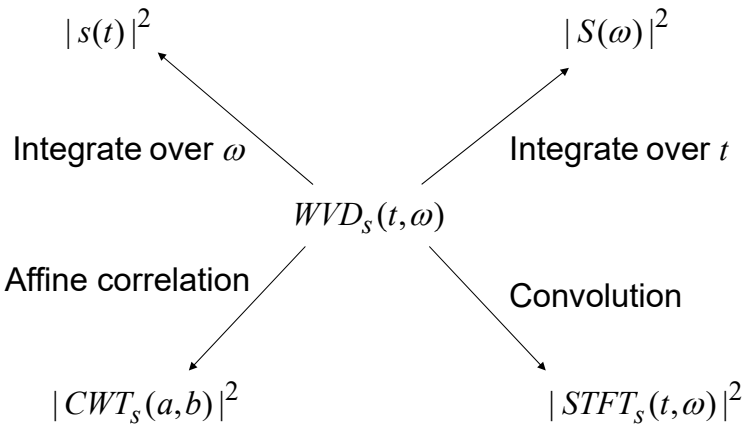
(Both follow immediately from the formulas for WVD)

WVD of multiple signals: Cross-terms

Wigner-Ville Distribution has many useful properties, and better resolution than STFT spectrogram. BUT
Applications are limited due to *cross-term interference.*

Consider composite signal $s(t) = s_1(t) + s_2(t)$. Then
$WVD_s(t,\omega) = WVD_{s_1}(t,\omega) + WVD_{s_2}(t,\omega) + 2\,\mathrm{Re}\left\{WVD_{s_1,s_2}(t,\omega)\right\}$
i.e. not only the sum of WVDs, but also the cross-term
$WVD_{s_1,s_2}(t,\omega)$
Also, cross-term is included at double the magnitude
of the auto-terms, so often obscures useful patterns.

From WVD to …?


$|s(t)|^2$
$|S(\omega)|^2$
Integrate over ω
Integrate over t
$WVD_s(t,\omega)$
Affine correlation
Convolution
$|CWT_s(a,b)|^2$
$|STFT_s(t,\omega)|^2$